Modeling repeated evolution in HSC population

Michael Tassia

March 10, 2026

Theorhetical modeling framework

Load libraries

```
library(tidyverse)
library(pbapply)
library(patchwork)
library(cowplot)
library(viridis)
```

Set parameters

Parameter values obtained from empirical estimates in Mitchell et al. 2022. Human genome size is a rough approximation.

```
hsc_pop <- c(80000,100000,120000)
ages <- seq(1,80,1)
annual_rates <- 2^seq(4,13)
diploid_genome_size <- 6.4e9
```

Run models

Code block will compute $P(X > 1)$ using Poisson and Binomial distributions (both results are identical). Note that these model the probability of a mutation occurring at the same site in any > 1 genomes in the population, disregarding the resulting state of the mutation.

```
## Poisson model
p.df <- pblapply(hsc_pop, function(x) {
  lapply(annual_rates, function(y) {
    df <- ppois(q = 1,
                lambda = ((y / diploid_genome_size) * x * ages),
                lower.tail = F) %>%
      as.data.frame() %>%
      rownames_to_column(var="age") %>%
      mutate(age = as.numeric(age)) %>%
      rename("p" = ".") # set the prob column name to p and convert rowname to age column

    df$N <- x # set HSC population size for df
    df$annual_muts <- as.factor(y) # set annual mutation rate for df
    df <- df %>% select(age, N, annual_muts, p)
    return(df)
  })
}) %>%
  bind_rows() %>%
```

```
  tibble()

## Binomial model
b.df <- pblapply(hsc_pop, function(x) {
  lapply(annual_rates, function(y) {
    df <- pbinom(q = 1,
                 prob = ((y / diploid_genome_size) * ages),
                 size = x,
                 lower.tail = F) %>%
      as.data.frame() %>%
      rownames_to_column(var = "age") %>%
      mutate(age = as.numeric(age)) %>%
      rename("p" = ".")

    df$N <- x
    df$annual_muts <- as.factor(y)
    df<-df %>% select(age, N, annual_muts, p)
    return(df)
  })
}) %>%
  bind_rows() %>%
  tibble()
```

Plot results

The plots below show curves as $P(X > 1)$ for the mean estimate for HSC population size, and ribbons show the range $\pm 20,000$ cells. Note that these are the conditional CDFs for the population and does not reflect sampling noise.

